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Optimal staging of reusable launch vehicles for minimum life cycle cost

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ABSTRACT

This paper proposes an optimal staging procedure of a reusable launch vehicle considering its life cycle operations.

Instead of minimizing the lift-off mass of a single launch adopted by traditional optimal sizing problems, the proposed procedure minimizes the total life cycle cost of reusable stages using an integrated optimization module and a cost model. The recurring/nonrecurring cost models are established to compute the life cycle cost of a reusable launch vehicle throughout the operational period of its reusable stage. The integrated optimization module obtains a vehicle configuration and ascent/descent trajectories to put the desired payload into the target orbit while minimizing the expendable stage mass for a given reusable stage. The validity and effectiveness of the proposed procedure are demonstrated through an optimal staging case study for a real reusable launch vehicle, Falcon 9.

1. Introduction

The demand for space transportation, whose fulfillment with the current launch capacity (using expendable vehicles) seems unrealistic, has continuously increased [1]. Therefore, a reusable launch vehicle (RLV) is an attractive option for cost- and time-effective space transportation. The saving in the manufacturing cost of the first stage and the reduction in the preparation period is significant enough to cover the cost for reuse (e.g., recovery, refurbishment, and additional propellant), which should be sufficiently low for the RLV to be a viable option. Vertical take-off vertical landing (VTVL), a flight-proven concept for RLVs, is efficient in terms of responsiveness and the cost incurred by reuse.

The optimal staging is an essential process during the conceptual design of a launch vehicle, which is directly connected to the launch cost. We can find the optimal mass configuration of a launch vehicle for a given mission requirement and propulsion/structural characteristics (e.g., required velocity increment, specific impulse, structural ratio) through the process. Various staging methods for expendable launch vehicles have been explored so far, including traditional approaches [2–14], methods combined with the optimal control theory [15,16], and procedures integrated with trajectory optimization for better accuracy / additional constraints [17–19]. On the other hand, the studies on the optimal staging of a reusable launch vehicle are relatively scarce, even though several countries and entities are developing or operating RLVs [20–23]. Recently, Dresia et al. developed an optimization framework for an RLV by combining the optimal staging module with a simplified velocity increment calculation [24]. Brevault et al. proposed a multi-objective multidisciplinary design optimization approach for an RLV with a sizing subroutine that interacts with propulsion, aerodynamics, and trajectory modules [25]. Jo and Ahn pointed out that previous staging methods for expendable launch vehicles are not suitable for RLVs in that the additional propellant required for soft landing is not used during the ascending phase. They proposed an optimal staging method for RLVs by expanding Koch's approach considering the velocity components during the descent phase [18,26].

The objective function of traditional optimal staging problems is to maximize the payload mass for a given lift-off mass (alternatively, to minimize the lift-off mass for a given payload mass). In an expendable launcher, we usually assume that the launch cost is proportional to the launch mass, and this objective function leads to cost minimization. However, reducing the lift-off mass of an RLV does not always decrease the total launch cost throughout its life cycle. The life cycle cost of an RLV is the sum of multiple expendable stage costs and a single reusable stage cost. Fig. 1 illustrates a situation where an RLV designed for the minimum staging method is not the same as the one with the optimized lift-off mass. We assumed that only the first stage is reusable, and the fairing/op-

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Nomenclature

a	semi-major axis
b	sensitivity factor for cost
C	cost
C_f	faring cost
C_{launch}	cost per launch
C_L^I	transformation matrix between the local and inertial frames
C_{total}	total life cycle cost
$\mathbf{D}$	drag force vector
e	eccentricity
$\mathbf{g}$	gravity acceleration vector
h	altitude
h_{rb}	reentry burn altitude
inc	inclination angle
J	objective function
k	constant value for the cost
Lat_{launch}	launch site latitude
$\mathrm{Lat}_{landing}$	landing site latitude
m	mass
$\dot{m}$	mass flow rate
m_L	stage payload mass
m_{pay}	payload mass
n	number of stages
n_r	number of reuses
p	learning factor
q	dynamic pressure
$\mathbf{r}$	position vector
$\mathbf{r}_{launch}$	position vector of a launch site
$\mathbf{r}_{landing}$	position vector of a landing site
R_E	radius of the Earth
t	time
t_{co}	coasting time
t_{ff}	free fall time
T	thrust (vacuum)
$\mathbf{u}$	control input
$\mathbf{v}$	velocity vector
v_{ex}	effective exhaust velocity (vacuum)
$\mathbf{v}_{launch}$	initial velocity vector of a launch site
$\mathbf{v}_{landing}$	velocity vector of a landing site
V_r	relative velocity
$V_{r,re}$	reentry relative velocity
$\mathbf{x}$	design variable
Δv	velocity gain
Δv_{loss}	velocity loss
Δv_{req}	ideal velocity increment
α	angle of attack
ε	structural ratio
κ	iteration value for staging algorithm
λ	payload ratio
θ	pitch angle
ω_E	rotation velocity of the Earth
ψ	yaw angle

Subscripts

0	initial value
a	value for the ascent phase
d	value for the descent phase
e	value for engine
f	final value
i	value of i^{th} stage
o	value for operation
p	value for propellant
r	value for reuse
s	value for structure

Superscripts

$*$	optimal value
(D)	value for development
l	value considering velocity loss
(O)	value for operation
(p)	phase number
(P)	value for production
T	target value

eration cost is the same for the two models. In the case of no reuse, the cost for the minimum lift-off mass is cheaper than that for the minimum life cycle cost. However, the situation changes as the number of reuses increases. A less expensive second stage saves $1.5 million at every launch. However, the more expensive first (reusable) stage is produced only once, making a less significant impact on lifecycle cost. This example implies that we can expect additional cost savings by designing the stages to minimize the life cycle cost.

Despite the importance of cost estimation in the design process of RLVs, studies concerning cost-optimal design for RLVs are scarce. Koelle emphasized the requirement of finding a cost-optimal configuration of RLVs and analyzed cost per payload mass utilizing a traditional cost estimation model for fully reusable launch vehicles [27]. However, applying the conventional cost estimation methods to RLVs is limited due to the lack of empirical data [28], necessitating the development of a new model adequate for reusable vehicles.

This paper proposes an optimal staging procedure that minimizes the life cycle cost of a VTVL reusable launch vehicle. Instead of separating the staging and trajectory optimization modules while communicating information required for design iteration as the conventional staging methods [18,26], this study developed an integrated module that simultaneously optimizes the trajectory and stage configuration to put the specified payload mass into a target orbit. A cost model that can assess the life cycle cost of a reusable launch configuration is established, and the optimization module adopts the life cycle cost (or cost per launch) as an objective function to minimize. The optimization module minimizes the life cycle cost for a given first (reusable) stage and payload. Fig. 2 illustrates the overall optimization procedure. The procedure begins with selecting the set of reusable stage mass values. Then, a cost model that can compare the relative cost among the various mass configurations corresponding to the set of reusable stage mass values is established. Afterward, the optimization module runs multiple times with the selected reusable stage mass set and the cost model, yielding associated optimal configurations and relevant launch costs. Finally, the procedure finds the optimal configuration for the minimum launch cost. More details on the proposed procedure are presented in Section 3.

The contribution of this study is twofold. First, an integrated optimization module for a reusable launch vehicle that can compute the optimal trajectory and upper-stage sizing for a given reusable stage specification is developed. A model that can estimate the life cycle cost is constructed and used as an objective function of the optimization module. The trajectory optimization involves both the ascending and landing flights with practical constraints. Second, a procedure for design optimization of a reusable

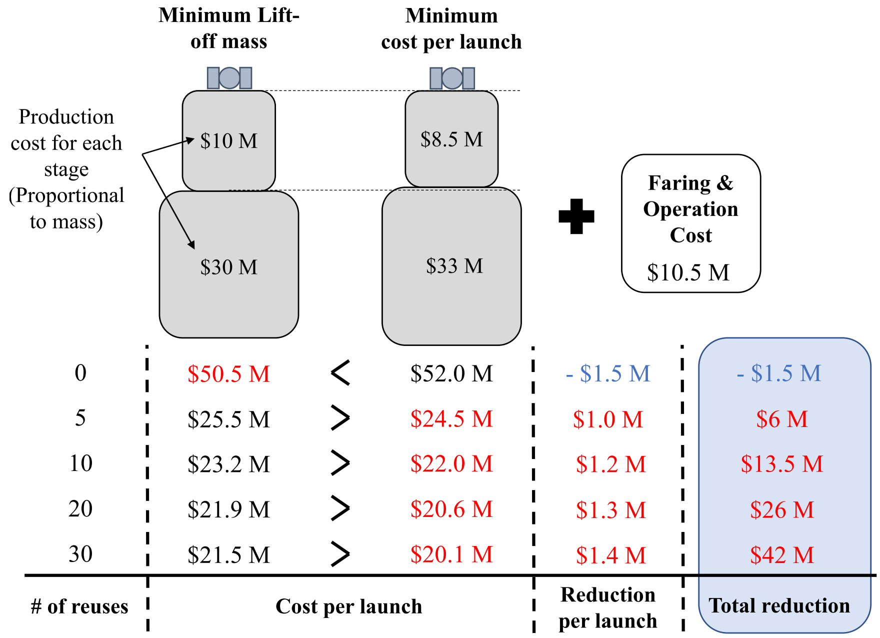

Fig. 1. Possible cost reduction of a partially reusable launch vehicle from minimum lift-off mass design.

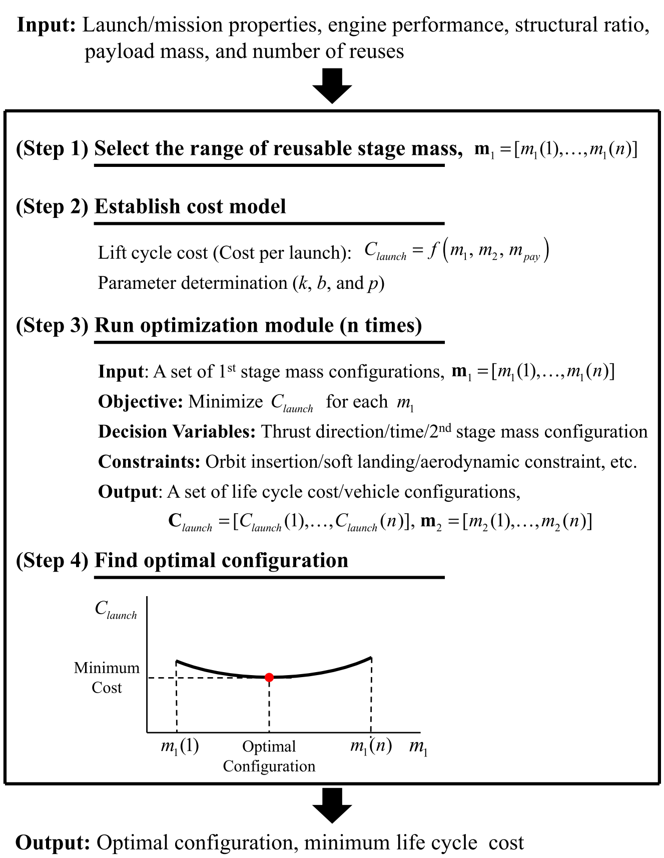

Fig. 2. A four-step procedure for design optimization of a reusable launch vehicle.

launch vehicle is proposed. The proposed four-step procedure provides the set of optimal vehicle configurations and associated cost values for various numbers of reuses. The validity and effectiveness of the proposed approach are demonstrated through a case study of an optimal stage design of a reusable launch vehicle with realistic parameters.

2. Review: optimal staging of a reusable launch vehicle for the minimum lift-off mass

This section reviews and improves the optimal staging method for the minimum lift-off mass of an RLV proposed by Jo and Ahn [26]. A traditional optimal sizing problem assumes the structural factor (ε_i) and the effective exhaust velocity defined in a vacuum condition ($v_{ex,i}$) for each stage as pre-determined parameters [18]. In addition, they introduced two mass ratios for the reusable stage (the first stage) as follows

$$\varepsilon_{a,1} = \frac{m_{s_a,1}}{m_{s_a,1} + m_{p_a,1}}, \tag{1}$$

$$\varepsilon_{d,1} = \frac{m_{s_d,1}}{m_{s_d,1} + m_{p_d,1}}. \tag{2}$$

In these equations, $m_{p_a,1}/m_{p_d,1}$ are the propellant mass values, adding up to become $m_{p,1}$ ($m_{p,1} = m_{p_a,1} + m_{p_d,1}$). Also, $m_{s_d,1}$ ($= m_{s,1}$) is the effective structural mass for the descent phase and $m_{s_a,1}$ is the effective structural mass for the ascent phase ($m_{s_a,1} = m_{s_d,1} + m_{p_d,1}$). Note that the (overall) structural factor for the reusable stage (ε_1) is the product of the ascent/descent phase structural factors expressed as

$$\varepsilon_1 = \frac{m_{s,1}}{m_{s,1} + m_{p,1}} = \varepsilon_{a,1} \cdot \varepsilon_{d,1}, \tag{3}$$

where $\varepsilon_{a,1}$ and $\varepsilon_{d,1}$ are defined in Eqs. (1)–(2). Fig. 3 provides the definitions of these mass values graphically.

We can express the velocity gain for the ascent phase of each stage ($\Delta v_{a,i}$) by using the payload ratio (λ_i) and the structural factor (ε_i) for the expendable ($i \neq 1$) and reusable ($i = 1$) stage as [18,26]

$$\Delta v_{a,i} = \begin{cases} v_{ex,i} \ln \frac{1+\lambda_i}{\varepsilon_i+\lambda_i} (= \Delta v_i) & (i \neq 1) \\ v_{ex,1} \ln \frac{1+\lambda_1}{\varepsilon_{a,1}+\lambda_1} & (i = 1) \end{cases}, \tag{4}$$

where λ_i is the ratio of i^{th} payload mass ($m_{L,i}$) to i^{th} stage mass ($m_{s,1} + m_{p,i}$) defined as

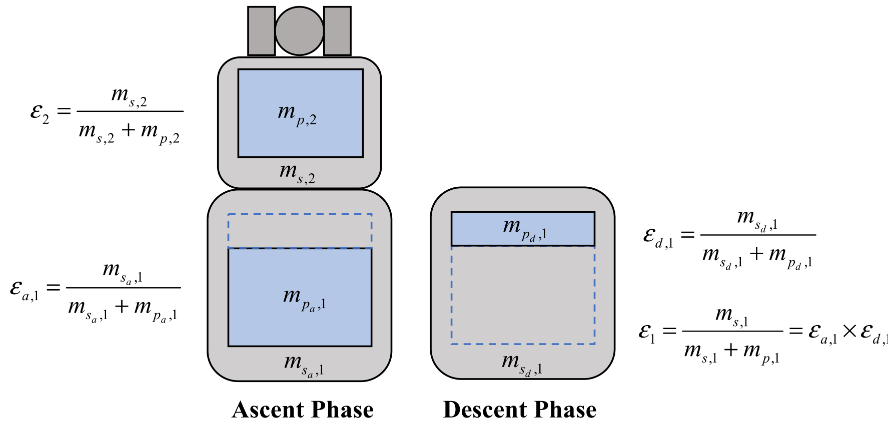

Fig. 3. Mass components and ratios for ascent/descent phases of a reusable stage.

$$\lambda_i = \frac{m_{L,i}}{m_{s,i}+m_{p,i}}. \tag{5}$$

In Eq. (5), $m_{L,i}$ is defined as

$$m_{L,i} = \begin{cases} m_{0,(i+1)} = \sum_{j=i+1}^{n}(m_{s,j}+m_{p,j}) + m_{pay} & (i<n) \\ m_{pay} & (i=n) \end{cases}, \tag{6}$$

where m_{pay} is the payload mass for the launch vehicle.

The optimal staging algorithm for the minimum lift-off mass consists of two steps. The first step finds the optimal distribution of the ideal velocity gain to each stage for the minimum lift-off mass. The second step calculates the mass configuration by adding the estimated velocity losses to the ideal velocity gains coming from the first part. The first step is formulated as the following optimization problem (problem **P**):

$$\max_{\boldsymbol{\lambda}} J(\boldsymbol{\lambda}) = \left(\frac{m_{pay}}{m_{0,1}}\right) = \prod_{i=1}^{n}\frac{\lambda_i}{1+\lambda_i}, \tag{7}$$

subject to

$$\begin{aligned} &\left(\Delta v_{a,1} + \sum_{i=2}^{n}\Delta v_i\right) - \Delta v_{req} \\ &= \left(v_{ex,1}\ln\frac{1+\lambda_1}{\varepsilon_{a,1}+\lambda_1} + \sum_{i=1}^{n}\left(v_{ex,i}\ln\frac{1+\lambda_i}{\varepsilon_i+\lambda_i}\right)\right) - \Delta v_{req} \\ &= 0. \end{aligned} \tag{8}$$

In Eq. (8), Δv_{req} is the ideal velocity increment to put the payload into the target orbit expressed as

$$\Delta v_{req} = v^T - v_{\text{launch}}, \tag{9}$$

where v^T is the target velocity obtainable from the target orbit information, and v_{launch} is the initial velocity of the launch vehicle resulting from the rotation of the Earth. Note that Eq. (9) is valid regardless of the launch direction since the ideal velocity increment is defined as the magnitude difference between two velocity vectors in this paper. The direction mismatch between initial and target velocity vectors is reflected as velocity losses.

The optimal solution of problem **P** is expressed as

$$\lambda_i^* = \begin{cases} \frac{\kappa\varepsilon_{a,1}}{(1-\varepsilon_{a,1})v_{ex,1}-\kappa} & (i=1) \\ \frac{\kappa\varepsilon_i}{(1-\varepsilon_i)v_{ex,i}-\kappa} & (i\neq 1) \end{cases}, \tag{10}$$

where superscript $*$ denotes the optimal value. A constant κ satisfies the following equation

$$\begin{aligned} \Delta v_{a,1} + \sum_{i=2}^{n}\Delta v_i &= \Delta v_1 - \Delta v_{d,1} + \sum_{i=2}^{n}\Delta v_i \\ &= v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} - v_{ex,1}\ln\frac{1}{\varepsilon_{d,1}} \\ &\quad + \sum_{i=2}^{n}\left(v_{ex,i}\ln\frac{v_{ex,i}-\kappa}{v_{ex,i}\varepsilon_i}\right) \\ &= \Delta v_{req}, \end{aligned} \tag{11}$$

where $\Delta v_{d,1}$ is the ideal velocity increment provided by the rocket for the descent phase. The value of $\Delta v_{d,1}$ for a soft landing is computed as

$$\begin{aligned} \Delta v_{d,1} &= \Delta v_{a,1} + v_{\text{launch}} - v_{\text{landing}} \\ &= \Delta v_{a,1} + R_E\omega_E\cos(\text{Lat}_{\text{launch}}) - R_E\omega_E\cos(\text{Lat}_{\text{landing}}), \end{aligned} \tag{12}$$

where R_E and ω_E are the radius and rotation velocity of the Earth. Also, $\text{Lat}_{\text{launch}}$ and $\text{Lat}_{\text{landing}}$ are the latitudes of the launch and landing sites, respectively. From Eqs. (11) and (12), $\Delta v_{a,1}$ can be expressed as

$$\begin{aligned} \Delta v_{a,1} &= v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} - v_{ex,1}\ln\frac{1}{\varepsilon_{d,1}} \\ &= v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} - \Delta v_{d,1} \\ &= v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} \\ &\quad - \left(\Delta v_{a,1} + R_E\omega_E\cos(\text{Lat}_{\text{launch}}) - R_E\omega_E\cos(\text{Lat}_{\text{landing}})\right) \end{aligned} \tag{13}$$

Rearranging Eq. (13) provides the expression for $\Delta v_{a,1}$ as

$$\begin{aligned} \Delta v_{a,1} = \frac{1}{2}\Bigg(& v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} - R_E\omega_E\cos(\text{Lat}_{\text{launch}}) \\ & + R_E\omega_E\cos(\text{Lat}_{\text{landing}})\Bigg). \end{aligned} \tag{14}$$

Putting Eq. (14) into Eq. (11) yields

$$\begin{aligned} \frac{1}{2}\Bigg(& v_{ex,1}\ln\frac{v_{ex,1}-\kappa}{v_{ex,1}\varepsilon_1} - R_E\omega_E\cos(\text{Lat}_{\text{launch}}) \\ & + R_E\omega_E\cos(\text{Lat}_{\text{landing}})\Bigg) \end{aligned}$$

$$+\sum_{i=2}^{n}\left(v_{ex,i}\ln\frac{v_{ex,i}-\kappa}{v_{ex,i}\varepsilon_i}\right)-\Delta v_{req}=0. \tag{15}$$

One can obtain the value of κ by solving Eq. (15) numerically. Then the optimal distribution of the ideal velocity increments for different stages/phases ($\Delta v^*_{a,1}$, $\Delta v^*_{d,1}$, and Δv^*_2) can be calculated by applying the κ value to Eq. (14), Eq. (12), and Eq. (4), respectively.

The second step considers various velocity loss values to adjust the "ideal" optimal sizing result obtained in the first step. The actual velocity increment values ($\Delta v_{a,i}$ and $\Delta v_{d,i}$) are expressed as

$$\Delta v_{a,i}=\Delta v^*_{a,i}+\Delta v_{loss_a,i}, \tag{16}$$

$$\Delta v_{d,1}=\Delta v^*_{d,1}+\Delta v_{loss_d,1}, \tag{17}$$

where $\Delta v_{loss_a,i}$ and $\Delta v_{loss_d,1}$ are the velocity losses during the ascent and descent phases, respectively. The optimal payload ratio considering velocity losses (λ_i^{l*}) can be calculated by combining Eqs. (4) and (16) as

$$\lambda_i^{l*}=\begin{cases}\dfrac{\varepsilon_i\cdot e^{\frac{\Delta v_i}{v_{ex,i}}}-1}{1-e^{\frac{\Delta v_i}{v_{ex,i}}}}=\dfrac{\varepsilon_i\cdot e^{\frac{\Delta v_i^*+\Delta v_{loss,i}}{v_{ex,i}}}-1}{1-e^{\frac{\Delta v_i^*+\Delta v_{loss,i}}{v_{ex,i}}}} & (i\neq 1)\\[2ex] \dfrac{\varepsilon_{a,1}\cdot e^{\frac{\Delta v_{a,1}}{v_{ex,1}}}-1}{1-e^{\frac{\Delta v_{a,1}}{v_{ex,1}}}}=\dfrac{\varepsilon_{a,1}\cdot e^{\frac{\Delta v_{a,1}^*+\Delta v_{loss_a,1}}{v_{ex,1}}}-1}{1-e^{\frac{\Delta v_{a,1}^*+\Delta v_{loss_a,1}}{v_{ex,1}}}} & (i=1)\end{cases} \tag{18}$$

Finally, we can find the optimal mass configuration by using the definition of the payload ratio expressed in Eqs. (5) and (6). Note that we use a vacuum effective exhaust velocity ($v_{ex,i}$), and its change with the altitude is reflected as a velocity loss term (pressure loss) in this study.

Koch [18] and Jo and Ahn [26] utilized the trajectory optimization module to predict the accurate velocity loss values. The trajectory optimization module uses the mass configuration coming from the optimal staging module and finds the maximum payload mass and associated velocity losses. The overall design procedure that iterates between the two modules continues until the variables linking the two modules (i.e., payload mass and velocity loss values) converge. Details on the optimal sizing of a reusable launch vehicle based on the iterations between the two modules are available in reference [26]. Note that the original procedure presented in the reference handles the ideal descent velocity gain ($\Delta v_{d,1}$) as an iteration variable obtainable from the trajectory optimization module. The authors later revised the procedure by introducing the analytic expression of $\Delta v_{d,1}$ (Eq. (12)), reducing the number of iteration variables with better convergence. This section presents the revised procedure to provide the readers with up-to-date information.

This procedure is valid for designing a minimum lift-off mass configuration of a reusable launch vehicle. The following section proposes a new approach that can optimize the cost-per-launch of a reusable launch vehicle considering its life cycle operations.

3. Optimal staging for minimum life cycle cost

The proposed optimal staging for minimum life cycle cost is a four-step procedure using an *optimization module* with the cost model depicted in Fig. 2. The range of first stage mass is selected, and a set of first stage mass values is defined within the range. The second step establishes the cost model used in the optimization module, and the life cycle cost (or cost per launch) can be calculated using the model. The model is applied to the optimization module as an objective function to minimize. It is essential to establish an appropriate cost model reflecting the operational characteristics of an RLV.

The third step runs the *optimization module* for a set of first-stage mass values within the range of interest. The *optimization module* combines the optimal sizing and the trajectory optimization procedure introduced in Section 2 as a single nonlinear optimization problem. The optimization module receives the first stage mass as an input. The outputs of the module are: 1) the optimal launch vehicle configuration (i.e., staging result), 2) the life cycle cost corresponding to the optimal configuration, and 3) the optimal trajectory (both ascent and landing). Thus, a set of optimal launch vehicle configurations and life cycle cost (output) associated with given reusable stage masses (input) are obtained after the third step.

The last (fourth) step generates the curves representing the cost per launch versus the reusable stage mass from the outputs of the third step. Then it selects the optimal (minimum cost) launch vehicle configuration based on the curves. The proposed optimal staging for minimum life cycle cost is a four-step procedure presented in Fig. 2. The following two subsections explain the cost model and optimization module in detail.

3.1. Parametric cost model for a reusable launch vehicle

This subsection introduces a parametric cost model using cost estimation relationships (CER) to assess the life cycle cost of a reusable launch vehicle used in this study. The cost model presented in this subsection can calculate the total launch cost corresponding to the mass configuration. The model categorizes the cost elements into development, production, and operational costs [29]. Estimating the accurate cost, especially the development cost, is a considerably complex process. Additionally, the available references and empirical data for reusable launch vehicles are scarce. It is important to note that the cost model in the staging process is not necessarily accurate. The goal of the model in the staging process is to compare the cost among the feasible configurations to find an optimal mass distribution that guarantees the minimum cost. In this study, the model evaluates the relative cost of the particular configuration to another. The cost model is established based on the following assumptions:

1) Launch cost is a function of mass only.
2) The development/production cost of the subsidiary elements required for the recovery (e.g., drone ship or landing site) is included in the first stage development/production cost.
3) All costs are expressed in the current fiscal year value.
4) Launch failure is not considered.

The launch cost of an RLV throughout its life cycle (C_{total}) and the cost per launch (C_{launch}) are defined as follows [30]

$$\begin{aligned}C_{total}=\;&C_{e,1}^{(D)}+C_{s,1}^{(D)}(m_{s,1})+C_{e,2}^{(D)}+C_{s,2}^{(D)}(m_{s,2})\\&+C_{e,1}^{(P)}+C_{s,1}^{(P)}(m_{s,1})\\&+\sum^{n_r+1}\left(C_{e,2}^{(P)}+C_{s,2}^{(P)}(m_{s,2})+C_f^{(P)}(m_{s,2})\right)\\&+\sum^{n_r}C_r^{(O)}(m_{s,1})+\sum^{n_r+1}C_l^{(O)}(m_0),\end{aligned} \tag{19}$$

$$C_{launch}=\frac{C_{total}}{n_r+1}, \tag{20}$$

where C_e, C_s, and C_f are the costs of the engine, structure, and fairing, respectively. The superscripts (D), (P), and (O) denote the development, production, and operation costs, respectively. The cost of the structure (C_s) includes all the subsystems required for the flight. Additionally, C_l is the launch operation cost (e.g., insurance fee, launchpad operation cost, delivery cost), and C_r is the additional operating cost required for reusing the first stage (e.g., refurbishment, storage/transportation of recovered stage, drone ship fuel/operations, and infrastructure). It is more reasonable to treat the fairing cost as the function of the payload in practice since the larger payload enforces a more spacious room inside the payload. However, in this paper, we assumed that the fairing cost depends on the second-stage structural mass to demonstrate that the proposed procedure can capture the difference in fairing mass. In Eq. (19), n_r is the number of reuses, and the recurring cost elements incurred at every launch/reuse should be summed up to calculate the total cost. The cost per launch (C_{launch}) is computed by dividing the total cost (C_{total}) by the number of launches $(1 + n_r)$. Although production cost of the first stage and the development cost are not incurred in every launch, the cost per launch reflects them as represented in Eq. (20).

The development costs of the whole vehicle and production costs related to the first stage are the nonrecurring costs. In contrast, the production costs of the fairing, second stage, and its engine and launch/reuse operation costs are recurring costs incurred during a life cycle. In expendable launch vehicles, the production costs for each stage and fairing, engine cost, and operational cost (e.g., insurance fee, delivery cost, propellant) are the major cost factors incurred in each launch. Particularly, the first stage accounts for most of the production costs due to its large portion of the mass distribution. Therefore, reusing the first stage eliminates the re-production process, as seen in Eq. (19), resulting in a significant reduction in launch cost.

Except for the engine costs ($C_{e,1}$ and $C_{e,2}$) whose characteristics are given in the staging problem, we assumed that the costs are the functions of mass only. Since the engine for each stage is already determined before the optimal staging, it is reasonable that the engine costs are given. The development, production, and operation costs of each element are defined as [29]

$$C^{(D)} = k^{(D)} m^{b^{(D)}}, \tag{21}$$

$$C^{(P)} = k^{(P)} m^{b^{(P)}} \cdot F^{(P)}, \tag{22}$$

$$C^{(O)} = k^{(O)} m^{b^{(O)}} \cdot F^{(O)}, \tag{23}$$

where k and b are the constant value and sensitivity factor for a specific system, respectively. In addition, m is the reference mass, and F is the cost reduction factor of the i^{th} launch tied to a learning factor (p) expressed as

$$F = i^{\log_2 p}, \quad (i = 1, \ldots, n_r + 1). \tag{24}$$

The conventional approach introduced in [29] applies various factors such as technical development status, quality, and team experience for better accuracy. It is generally used to estimate the cost of a new launch vehicle from its mass properties based on empirical data, but the empirical data for RLVs is scarce. Note that we can find the cost-minimum configuration in the proposed procedure by comparing the relative cost among various mass configurations. Therefore, in the design process of a new launch vehicle utilizing the proposed approach, the parameters (k, b, and p) should be determined from the analysis that can represent cost differences depending on the mass variation of a target launch vehicle whose system basis is already designed instead of using the conventional values. Considering the possibility of the public subsidies and the depreciation of the development cost over launch is the potential future work as those factors are important, especially for the private companies, in planning a design of a new RLV.

3.2. Integrated optimization module

The *integrated optimization module* produces the launch vehicle configuration and associated trajectory to minimize the life cycle cost for a given reusable stage mass value. The module combines the optimal sizing and the trajectory optimization as a single nonlinear optimization problem, which is different from the framework introduced in Section 2. The proposed trajectory optimization finds the sizing result for a given reusable stage mass and payload mass, whereas the one described in Section 2 optimizes the payload mass for a given configuration.

The module considers two landing scenarios: 1) return to launch site (RTLS) and 2) downrange landing (DRL) on a drone ship. The reusable stage executes three burns during the descent phase with a specific purpose for each burn: 1) boost-back burn for changing the instantaneous impact point (IIP) [36] to the landing site (optional in DRL scenario), 2) reentry burn for decelerating to avoid damage on the rocket in the dense atmospheric area, and 3) terminal burn for reducing the residual velocity to zero for soft landing [31,32].

The flight sequence of an RLV consists of 8 distinct segments as shown in Fig. 4: two segments for ascending (first stage burn and second stage burn) and six for descending (coast after separation, boost-back burn, coast after boost-back burn, reentry burn, descent gravity turn, and terminal burn). We parameterized pitch/yaw angles defining the thrust direction during the thrusting and gravity turn segments (referred to as the optimal control phase in this paper). The pitch/yaw angles, in addition to the variables defining the expendable stage configuration, are the design variables of the integrated optimization problem.

Phases 1 and 2 are associated with the ascent trajectory. Phase 1 starts at the lift-off and terminates at the first stage's main engine cut-off (MECO). Phase 2 coincides with the burning period of the expendable stage. The trajectory analysis used in this study does not consider the fairing separation, and we assume that the second-stage structure mass ($m_{s,2}$) includes the faring mass. The decent trajectory comprises four phases: boost-back burn (Phase 3), reentry burn (Phase 4), descent gravity turn (Phase 5), and terminal burn (Phase 6). Phase 3 starts after the coasting of the reusable stage, and Phase 4 starts when the reusable stage reaches a pre-determined altitude. Phase 5 is initiated when the velocity of the reusable stage is decreased to a target value. Finally, Phase 6 starts with the free initial condition and terminates when the soft landing is completed. An optimal control formulation for the integrated optimization module is presented [33,34].

$$\min_{\mathbf{u}(t),\mathbf{x}} J\big(\mathbf{u}(t), \mathbf{x}\big) = C_{launch}, \tag{25}$$

subject to

(Design variables)

$$\mathbf{u}(t) = \left[\theta^{(p)}(t), \psi^{(p)}(t)\right] \quad (p = 1, \ldots, 6), \tag{26}$$

$$\mathbf{x} = \left[t_f^{(p)}, m_{s,2}, m_{p,2}\right] \quad (p = 1, \ldots, 6), \tag{27}$$

(Dynamic constraints)

$$\dot{\mathbf{r}} = \mathbf{v}, \tag{28}$$

$$\dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}) + \frac{\mathbf{D}(\mathbf{r}, \mathbf{v})}{m} + C_L^I T \left[\begin{array}{ccc} \cos\psi\cos\theta & \sin\psi & -\cos\psi\sin\theta \end{array}\right]^T, \tag{29}$$

$$\dot{m} = -\frac{T}{v_{ex}}, \tag{30}$$

(Path constraints)

$$\left|\alpha^{(1)}(t)\right| \le \alpha_{\max}^{(1)}, \tag{31}$$

$$q^{(1)} \le q_{\max}, \tag{32}$$

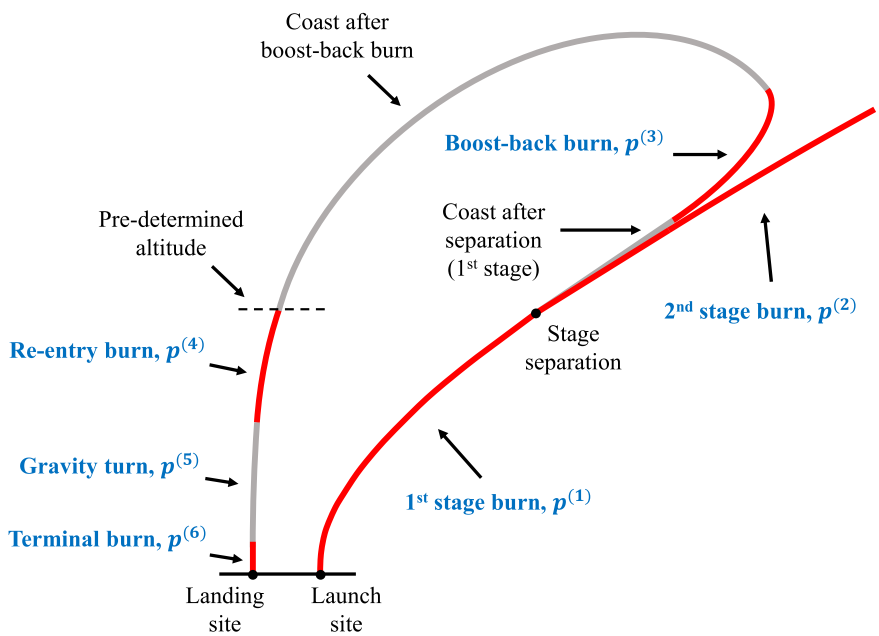

Fig. 4. Launch sequence and phases for a reusable launch vehicle.

$$\left|\alpha^{(5)}(t)-\pi\right| \leq \alpha_{\max}^{(5)}, \tag{33}$$

(Boundary conditions)

$$\left[\mathbf{r}_0^{(1)}, \mathbf{v}_0^{(1)}\right]=\left[\mathbf{r}_{\text{launch}}, \mathbf{v}_{\text{launch}}\right], \tag{34}$$

$$\left[a_f^{(2)}, e_f^{(2)}, inc_f^{(2)}\right]=\left[a^T, e^T, inc^T\right], \tag{35}$$

$$\left[\mathbf{r}_f^{(6)}, \mathbf{v}_f^{(6)}, m_f^{(6)}, \theta_f^{(6)}\right]=\left[\mathbf{r}_{\text{landing}}, \mathbf{v}_{\text{landing}}, m_{s,1}, \frac{\pi}{2}\right], \tag{36}$$

(Endpoint constraints)

$$\begin{aligned} &\frac{m_f^{(2)}-m_{pay}}{m_0^{(2)}-m_{pay}}=\frac{m_{s,2}}{m_{s,2}+m_{p,2}}=\varepsilon_2, \\ &m_0^{(2)}-m_f^{(2)}=\dot{m}^{(2)}\left(t_f^{(2)}-t_0^{(2)}\right), \end{aligned} \tag{37}$$

$$m_f^{(1)}=m_0^{(2)}+m_0^{(3)}, \qquad t_f^{(1)}\dot{m}^{(1)}+m_0^{(3)}=m_{p,1}+m_f^{(6)}, \tag{38}$$

$$\left[\mathbf{r}_0^{(3)}, \mathbf{v}_0^{(3)}\right]=f_1\left(\mathbf{r}_f^{(1)}, \mathbf{v}_f^{(1)}\right), \qquad t_0^{(3)}=t_f^{(1)}+t_{co}, \tag{39}$$

$$\left[\mathbf{r}_0^{(4)}, \mathbf{v}_0^{(4)}\right]=f_2\left(\mathbf{r}_f^{(3)}, \mathbf{v}_f^{(3)}\right), \qquad t_0^{(4)}=t_f^{(3)}+t_{ff}, \qquad h_0^{(4)}=h_{rb}, \tag{40}$$

$$V_{r,f}^{(4)} \leq V_{r,re}. \tag{41}$$

Eq. (25) presents the objective function of the problem: minimizing the cost per launch equivalent to minimizing the expendable stage mass according to the cost model in Eq. (19) in this paper. The design variables expressed in Eq. (26) and Eq. (27) are the histories of the pitch/yaw angles during the thrusting phases, the final time of each phase, and the expendable stage configuration. Superscript p denotes the phase number. Eqs. (28)–(30) describe the dynamics of RLVs defined in an Earth-Centered Inertial (ECI) frame where $\mathbf{r}$ and $\mathbf{v}$ are the position and velocity vectors, respectively, $\mathbf{g}(\mathbf{r})$ is the gravity acceleration vector, and $\mathbf{D}(\mathbf{r}, \mathbf{v})$ is the drag force vector. The spherical Earth model and U.S. standard atmosphere model are used for the dynamic equations. T and v_{ex} are respectively the thrust and effective exhaust velocity defined in a vacuum condition, which are given parameters. Also, C_L^I is the transformation matrix between the local and inertial frames.

Eq. (31) and Eq. (33) impose the constraint on the angle of attack during the ascent and the descent trajectories, respectively. Eq. (32) constrains the dynamic pressure of the vehicle within a pre-specified value, where dynamic pressure q is defined as

$$q=\frac{1}{2}\rho v_{rel}^2. \tag{42}$$

In this equation, ρ and v_{rel} are the air density and velocity of the launch vehicle relative to the Earth, respectively.

Eqs. (34)–(36) present the boundary conditions specified based on the launch/landing sites and target orbit information. Note that the subscripts 0 and f in these equations indicate the initial and final values, respectively. The RLV is launched at the launch site ($\mathbf{r}_{\text{launch}}$) with the velocity ($\mathbf{v}_{\text{launch}}$) obtained by the Earth's rotation. The target semi-major axis (a^T), eccentricity (e^T), and inclination angle (i^T) are the target orbital elements the expendable stage(s) should satisfy at the end of Phase 2 (orbit insertion). The reusable stage lands on the landing site ($\mathbf{r}_{\text{landing}}$) vertically with the inertial velocity of the landing site ($\mathbf{v}_{\text{landing}}$). The mass becomes identical to the structural mass of the reusable stage ($m_{s,1}$) at the end of Phase 6, where the propellant is fully consumed.

Eqs. (37)–(38) describe the endpoint constraints imposed to reflect the mass properties and discontinuity characteristics. Eq. (37) defines the relationship of the initial and final masses in Phase 2: the structural ratio of the second stage is restricted to the given value (ε_2), and the difference between the initial and final mass ($m_0^{(2)}-m_f^{(2)}$) is defined as the propellant mass consumed by the propulsion during this phase. Additionally, Eq. (38) specifies the mass property of the reusable stage. The mass before separation ($m_f^{(1)}$) equals the sum of the masses of the separated stages ($m_0^{(2)}, m_0^{(3)}$) at $t=t_f^{(1)}$. Also, the sum of the propellant mass consumption during Phase 1 ($t_f^{(1)}\dot{m}^{(1)}$) and the first stage mass right after separation ($m_0^{(3)}$) is identical to the first stage mass ($m_{p,1}+m_f^{(6)}=m_{p,1}+m_{s,1}$). The discontinuity occurs between Phase 1 and Phase 3 since the coasting element after the stage separation is not considered as the phase for the optimization, which is linked based on Lagrange's f and g functions (Eq. (39)) [35]. Eq. (40) explains the vehicle's state at the beginning of Phase 4

Table 1
Launch and reentry event sequence and constraints for case study [26,39].

Phase	Event	Event time, s	Endpoint constraint	Path constraint
$p^{(1)}$	S1 engine ignition	0	Eq. (34)	$\theta = 90$ deg
	S1 gravity turn begins	15	–	Eq. (31), $\alpha_{\max}^{(1)} = 1$ deg
	S1 engine cut-off & separation	$t_f^{(1)}$	–	
$p^{(2)}$	S2 engine ignition	$t_0^{(2)}(=t_f^{(1)})$	–	–
	S2 engine cut-off	$t_f^{(2)}$	Eq. (35)	–
$p^{(3)}$	S1 boost-back burn begins	$t_0^{(3)}$	Eq. (39), $t_{co} = 20s$	–
	S1 boost-back burn terminates	$t_f^{(3)}$	–	–
$p_0^{(4)}$	S1 reentry burn begins	$t_0^{(4)}$	Eq. (40), $h_{rb} = 60$ km	–
	S1 reentry burn terminates,	$t_f^{(4)}$	Eq. (41), $V_{r,f}^{(4)} = 800$ m/s	–
$p_0^{(5)}$	S1 gravity turn begins	$t_0^{(5)}(=t_f^{(4)})$	–	Eq. (33), $\alpha_{\max}^{(5)} = 2$ deg
	S1 gravity turn terminates	$t_f^{(5)}$	–	
$p^{(6)}$	S1 terminal burn begins	$t_0^{(6)}(=t_f^{(5)})$	–	–
	S1 landing	$t_f^{(6)}$	Eq. (36)	–

(reentry burn) when the altitude of the reusable stage becomes a predesigned value (h_{rb}). The initial time of the phase is the time after the free-fall flight time (t_{ff}) of the final time of Phase 3. One can obtain the free-fall flight time and the free-fall position at altitude h_{rb} using the procedure presented in reference [36]. The velocity vector at the given radius ($r_p = R_E + h_{rb}$) is derived as

$$\mathbf{v}_p = v_p\left(-\sin(\gamma_p)\cdot \mathbf{i}_p + \cos(\gamma_p)\cdot \mathbf{i}_t\right), \tag{43}$$

where v_p, γ_p are the magnitude of velocity and flight path angle at r_p, respectively, expressed as

$$v_p = \sqrt{v_0^2 - 2\frac{\mu}{r_0} + \frac{\mu}{r_p}}, \tag{44}$$

$$\gamma_p = \cos^{-1}\left(\frac{h_0}{r_p v_p}\right). \tag{45}$$

In Eqs. (44) and (45), v_0, r_0, h_0, and μ are the magnitudes of the current velocity, position and angular momentum vectors, and Earth gravitational parameter, respectively. Also, $\mathbf{i}_p$ and $\mathbf{i}_t$ in Eq. (43) are the unit vectors for position/tangential directions whose details are presented in [36].

The reusable stage re-enters the dense atmospheric area with high velocity in the descent phase, experiencing a large aerodynamic load. Therefore, decelerating the reusable stage before the reentry is necessary to prevent damage, which is translated as a constraint at the end of Phase 4 expressed in Eq. (41). h_{rb} and $V_{r,re}$ should be determined based on the analysis of an aerodynamic load (dynamic pressure and heat flux) during the reentry.

4. Case study

This section discusses case study results to demonstrate the validity and effectiveness of the proposed algorithm. We selected Falcon 9, a two-stage partial reusable launch vehicle developed by Space X, as the case study model. Also, the CRS-10, a resupply service to the international space station (ISS), was selected as the target mission for the case study [26,37]. The staging result obtained using the proposed algorithm is compared to the original vehicle specification and the conventional optimal staging result to minimize the lift-off weight. Table 1 exhibits the launch/reentry event sequence and constraints considered for the case study. We implemented the trajectory optimization module in MATLAB/GPOPS-II using pseudospectral optimization with the SNOPT nonlinear programming (NLP) solver [38]. The drag coefficient profile used for the case study came from reference [26].

Table 2
Cost parameter used for case study.

Parameters		Stage 1 structure	Stage 2 structure	Fairing	Launch	Reuse
Development	$k^{(D)}$	1.51	0.5	0.89	–	–
	$b^{(D)}$	0.5	0.55	0.45	–	–
Production	$k^{(P)}$	0.052	0.03	0.022	–	–
	$b^{(P)}$	0.6	0.65	0.65	–	–
Operation	$k^{(O)}$	–	–	–	0.0005	0.0004
	$b^{(O)}$	–	–	–	0.67	0.7

Table 3
Learning factor for recurring cost used for case study.

Recurring cost element	Stage 2 structure	Stage 2 engine	Fairing	Launch operation	Reuse operation
p	0.95	0.95	0.95	0.9	1.1

We assume that the first stage separation and the second stage engine ignition take place simultaneously. The first stage executes the boost-back burn using three out of nine engines 20 seconds after the coasting starts (t_{co}). After the termination of the boost-back burn, the vehicle is subject to another coasting before it reaches the predesigned altitude (h_{rb}) of 60 km, where the reentry burn begins with one engine. The thrust level of the final landing burn is half of a single engine's maximum thrust level. Note that the unspecified parameters representing event time in the Table are the design values to be optimized using the integrated optimization module.

Table 2 provides the cost parameters used for this study. We first determined the sensitivity factors (b) by referring to the empirical data commonly used for an expendable launch vehicle [29,40,41]. Afterward, the constant value (k) is selected based on the cost estimation of the original (existing) launch vehicle. The elements of the production and operation costs are estimated based on the data in references [42–46], public statements of Space X executives, and several assumptions (e.g., expected number of launches, gross margin). The total development cost can be found in [47], and it is distributed to the respective element in the development cost based on its portion of the production cost. Due to the lack of cost information, the cost values provided in this paper are rough approximate values from limited available resources. The unit of the cost and mass used to determine the cost parameters are *million USD* and kg, respectively. Table 3 presents learning factors for recurring costs. Traditional learning factors are used for

Table 4
Lift-off mass and cost per launch for various $m_{s,1}$ and n_r values.

Mass configuration, Mg			Cost per launch, *million USD*					
$m_{s,1}$	$m_{s,2}$	m_0	$n_r = 0$	$n_r = 10$	$n_r = 30$	$n_r = 50$	$n_r = 100$	$n_r = 200$
22.6	5.77	574.6	444.95	56.68	30.52	24.55	19.64	16.80
22.2[a]	5.50[a]	561.2	439.70	55.80	29.95	24.05	19.21	16.41
21.4	5.43	544.2	434.32	55.15	29.62	23.80	19.01	16.24
21.3	5.43	542.3	433.76	55.10	29.60	23.78	19.00	16.23
21.2	5.44	540.4	433.24	55.05	29.58	23.77	19.00	16.23
21.1	5.45	538.6	432.75	55.01	29.57	23.77	19.00	16.23
21.0	5.45	536.8	432.20	54.96	29.55	23.75	**18.99**	**16.23**
20.9	5.46	535.0	431.72	54.92	29.54	23.75	18.99	16.23
20.8	5.47	533.3	431.25	54.89	29.53	**23.75**	18.99	16.24
20.7	5.49	531.6	430.81	54.86	**29.53**	23.75	19.00	16.24
20.6	5.51	530.1	430.46	54.85	29.54	23.77	19.02	16.26
20.5	5.52	528.5	430.03	54.82	29.54	23.77	19.03	16.27
20.4	5.54	527.0	429.66	**54.81**	29.55	23.79	19.04	16.29
20.2[b]	5.60[b]	524.3	429.10	54.82	29.59	23.84	19.09	16.34
20.0	5.70	522.4	428.99	54.94	29.72	23.96	19.21	16.45
19.9	5.75	**521.3**	**428.84**	54.98	29.76	24.00	19.26	16.50
19.85	5.80	521.6	429.17	55.08	29.85	24.08	19.33	16.57
19.82	5.86	522.0	429.53	55.18	29.93	24.16	19.40	16.63

Note: Values in **underlined bold font** represent the minimum in each column.
[a] Original configuration.
[b] Identical to the result obtained from the traditional staging method presented in [26].

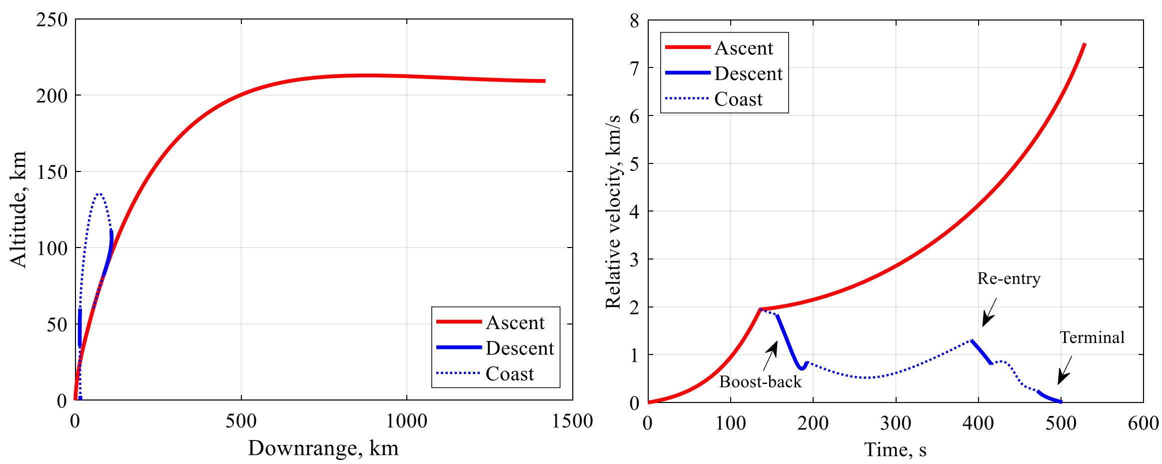

Fig. 5. Optimized trajectory with original configuration: (Left) altitude vs. downrange, (Right) velocity profile.

the structure and engine of stage 2, fairing, and launch operation costs. In contrast, we chose a value of more than 1 for the operation cost required for the recovery, which increases the cost as the launch continues. Considering that the number of components that require inspection, repair, and refurbishment increases with repeated recovery, it is reasonable to assume the learning factor of the operation cost for recovery is greater than 1.

4.1. Trajectory optimization with the original configuration

Before starting the optimal staging, we obtained an optimal trajectory with the original vehicle configuration to set up the baselines. Table 5 summarizes the vehicle configuration used for this procedure. The objective function of trajectory optimization is to maximize the payload mass (max $J = m_{pay}$). The constraints and boundary conditions presented in Eqs. (28)–(41) (except the dynamic pressure constraint presented in Eq. (32)) are imposed. Fig. 5, Table 5, and Table 6 present the results of the trajectory optimization with the original configuration.

The optimized payload mass (14.97 Mg) and the maximum dynamic pressure during the first phase (35 kPa) are used as input parameters ($m_{pay}, q_{\max}$) in the trajectory optimization module of the proposed staging method. The maximum dynamic pressure is an essential constraint in the optimal staging problem. The lower lift-off mass may lead to a higher thrust-to-weight ratio (T/W_0), increasing the maximum dynamic pressure during the first phase.

4.2. Optimal staging results

We conducted the optimal staging following the steps described in Section 3 and compared the results with the results of Section 2. The first stage mass considered in the procedure ranges between 19.82 Mg and 22.6 Mg. Table 4 exhibits the optimal second-stage structural mass, lift-off mass, and cost per launch for various $m_{s,1}$ and number of reuses. Note that $m_{s,2}$ in the second column is the optimized (minimum) second-stage structural mass obtained through the integrated optimization module for the corresponding $m_{s,1}$. The propellant mass for each stage ($m_{p,1}$ and $m_{p,2}$) can be easily calculated by combining the obtained structural mass and the structural factor values (ε). The cost per launch for the original configuration (the second row in Table 4) is identical to the one estimated from the references since the cost parameters are determined from the estimated cost for the original configuration.

The mass configuration for the minimum cost varies depending on the desired number of reuses. We can observe that the minimum lift-off mass does not guarantee the minimum launch cost

Table 5
Case study results ($n_r = 100$).

Category	Parameter	Original configuration	Min. Lift-off weight configuration	Conventional optimal staging [26]	Proposed method
Stage configuration, Mg	m_{pay}	14.97	(same as original)		
	m_0	**561.2**	**521.4**	**524.3**	**536.38**
	$m_{s,1}, m_{p,1}$	22.2, 411.0	19.9, 335.1	20.2, 374.0	21.0, 349.0
	$m_{s,2}, m_{p,2}$	5.50, 107.5	5.75, 112.3	5.60, 109.5	5.46, 106.6
Thrust (vacuum), *kN*	T_1, T_2	8225.8, 933.6	(same as original)		
Exit velocity (vacuum), m/s	$v_{ex,1}, v_{ex,2}$	3050, 3412	(same as original)		
Structural factor, %	$\varepsilon_1, \varepsilon_2$	5.125, 4.867	(same as original)		
	$\varepsilon_{a,1}, \varepsilon_{d,1}$	15.42, 33.24	13.72, 37.36	14.13, 36.28	14.82, 34.57
Cost, million USD	C_{launch}	**19.21**	**19.26**	**19.09**	**18.99**
	C_{total}	**1940.16**	**1944.96**	**1928.40**	**1917.72**

Table 6
Flight sequence comparison.

Phase	Event	Time, *s*			
		Original configuration	Min. lift-off weight configuration	Conventional optimal staging	Proposed method
$p^{(1)}$	S1 engine ignition	0	0	0	0
	S1 gravity turn begins	15	15	15	15
	S1 engine cut-off & separation	135.8	124.2	125.5	129.4
$p^{(2)}$	S2 engine ignition	135.8	124.2	125.5	129.4
	S2 engine cut-off	528.8	534.8	525.6	518.8
$p^{(3)}$	S1 boost-back burn begins	155.8	144.2	145.5	149.4
	S1 boost-back burn terminates	192.3	172.1	175.2	182.4
$p^{(4)}$	S1 reentry burn begins	391	366.4	371.3	380.4
	S1 reentry burn terminates,	415.5	383.6	389.7	401.6
$p^{(5)}$	S1 gravity turn begins	415.5	383.6	389.7	401.6
	S1 max. dynamic pressure (kPa)	444.4 (51.3)	418.7 (49.6)	423.8 (49.6)	433.2 (50)
	S1 max. heat rate (kW/m^2)	436.8 (22.1)	411.4 (24.6)	416.9 (24)	425.4 (23)
	S1 max. axial acceleration (g)	444.4 (4.0)	419.1 (4.4)	424.2 (4.3)	433.6 (4.2)
	S1 gravity turn terminates	471.3	454.7	458.6	465.1
$p^{(6)}$	S1 terminal burn begins	471.3	454.7	458.6	465.1
	S1 landing	501.5	475.7	480.7	490.1

regardless of the number of reuses, supporting the necessity of the proposed algorithm. Due to the extremely high development cost, the cost per launch for an expendable launch case ($n_r = 0$) is much higher than in the case of multiple reuses. However, the cost per launch decreases as n_r increases because of the learning effect and the nonrecurring costs divided by n_r. Fig. 6 presents the optimized (minimum) second-stage structural mass ($m^*_{s,2}$) versus the first-stage structural mass ($m_{s,1}$) and the optimized first/second stage masses versus the number of reuses. No feasible mass configurations exist under the minimum (optimized) structural mass curve presented in the figure. The optimal lift-off mass and associated cost (per launch) can be obtained by combining the stage mass components and cost coefficients. It is shown that the second stage mass varies in a narrow range (5.43 Mg to 5.86 Mg) for a given range of the first stage mass, leading to the slight cost variation corresponding to each configuration. It is because of the small structural factors of Falcon 9. Greater structural factors lead to a wider range of variations in the second stage mass and cost, emphasizing the importance of the proposed procedure.

Fig. 7 plots the cost per launch versus $m_{s,1}$ for several n_r values. Note that even with the reuse of the first stage, the second stage should be re-produced for every launch. Therefore as n_r increases, the launch cost depends more on $m_{s,2}$ than m_0. As a result, the optimal design point for a large n_r in Fig. 7 becomes closer to the case with the minimum $m_{s,2}$ in Fig. 6. Fig. 8 plots lift-off mass (m_0) and cost per launch versus first-stage structural mass ($m_{s,1}$) for an expendable vehicle case ($n_r = 0$). This proves that the conventional staging method for an RLV in [26] can find a near-optimal solution, and the launch cost for an expendable case is proportional to the lift-off mass.

The optimal mass configuration for the desired or appropriate number of reuses can be determined based on the optimization result. Table 5 presents the case study results comparing the original configuration, the minimum-lift-off staging [26], and the proposed method for $n_r = 100$. By choosing a lighter second stage, the proposed staging method saves 0.22/0.11/0.27 million USD per launch (total 22.45/10.68/27.25 million USD) during the 101 launches compared to the original/minimum lift-off weight/conventional configuration, respectively. This study result presents the potential of the proposed method in cost reduction. Fig. 9 shows the flight profile of the minimum-cost configuration RLV, showing a slight difference from the trajectory with the original configuration presented in Fig. 5. Table 6 exhibits the flight sequence for three vehicle configurations (original, minimum lift-off mass, and minimum cost).

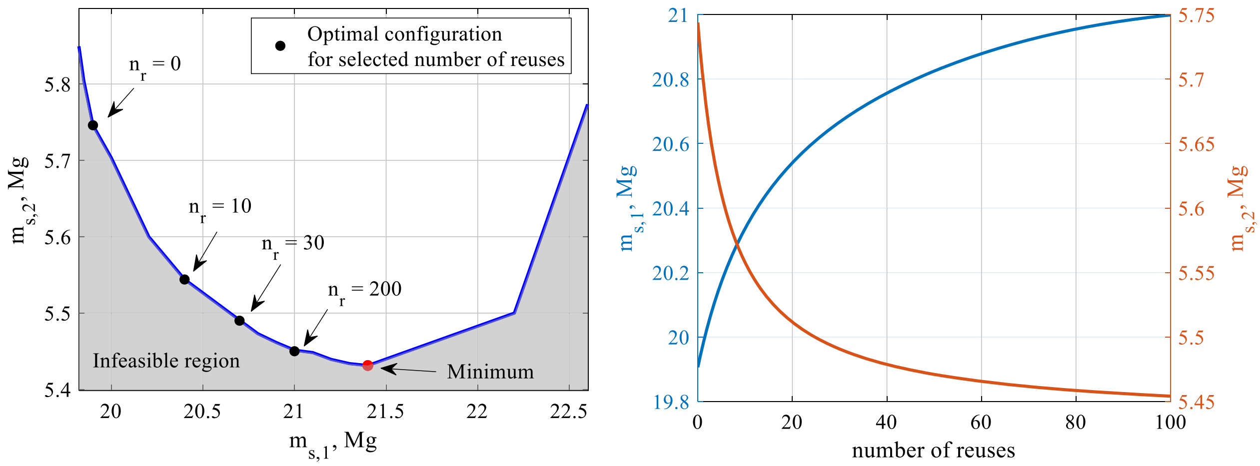

Fig. 6. (Left) Minimum (optimized) second-stage structural mass ($m^*_{s,2}$) versus first-stage structural mass ($m_{s,1}$), (Right) Minimum (optimized) first/second stage structural masses versus the number of reuses (n_r).

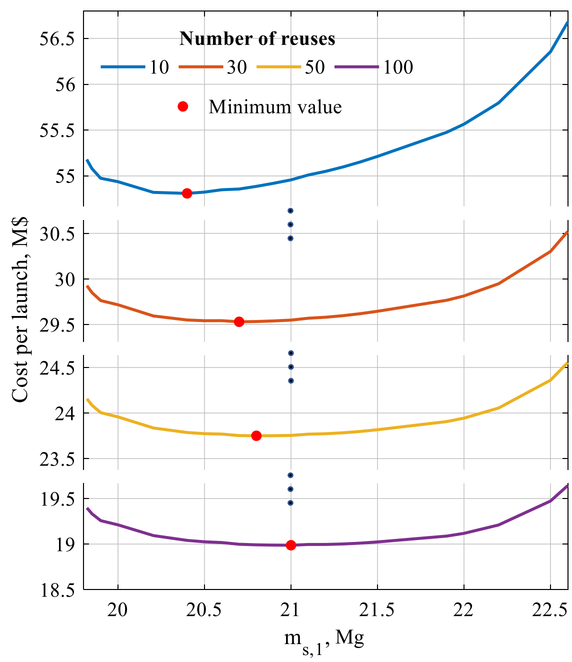

Fig. 7. Cost per launch versus $m_{s,1}$ for several n_r values.

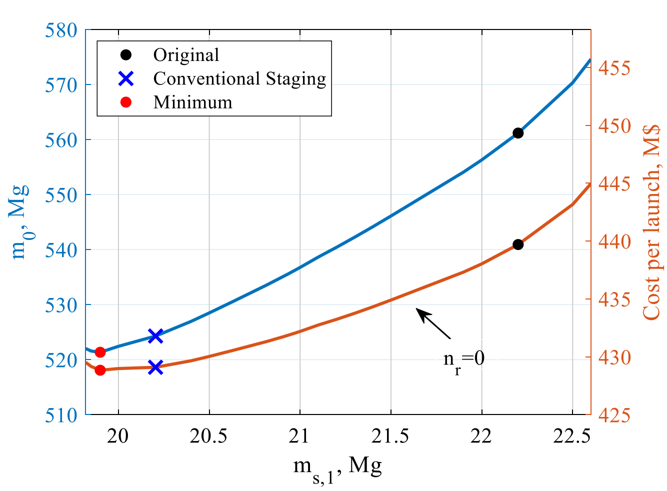

Fig. 8. Lift-off mass and cost per launch versus $m_{s,1}$ ($n_r = 0$).

Even though the proposed staging procedure for RLVs is validated through a case study on Falcon 9, more case studies should be carried out to figure out limitations and demonstrate that the procedure can be applied generally. Notably, the additional case study on the DRL scenario is an essential process for RLVs designed to perform two scenarios (RTLS and DRL) according to the mission in that the optimal configurations for both scenarios may be different. The RTLS requires more propellant for soft landing than the DRL, resulting in lower payload mass. Therefore, a trade-off study between optimal configurations for the two scenarios, based on the number of launches for each scenario, needs to be conducted to find the global optimal configuration that can efficiently cover both scenarios.

5. Conclusions

A new optimal staging procedure of a reusable launch vehicle considering its life cycle operations is proposed. The new procedure minimizes the total cost of a reusable vehicle throughout its life cycle using an integrated optimization module and the cost model. The validity and effectiveness of the proposed algorithm are demonstrated through an optimal staging case study using an actual reusable vehicle (Falcon 9). The case study result proves that the minimum lift-off mass configuration obtainable by a conventional stage optimization does not necessarily result in the minimum life cycle cost, justifying the benefit of the proposed approach. Refinement of the cost and mass models reflecting practical consideration (e.g., launch risk for a large number of reuses, separate consideration of engine/fairing masses, and development cost depreciation) and additional case studies on a DRL scenario or another RLV are potential subjects for future study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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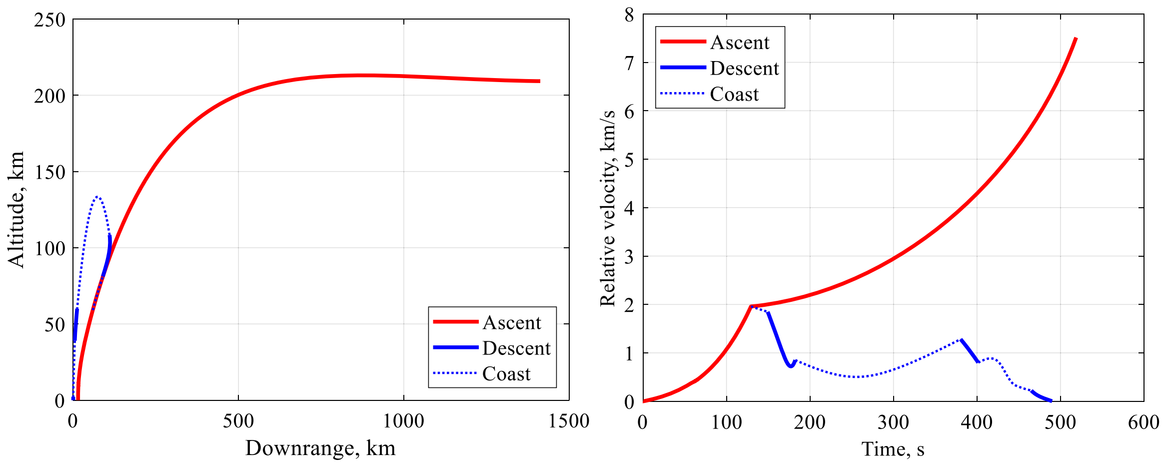

Fig. 9. Flight profile for minimum cost configuration: (Left) altitude versus downrange, (Right) velocity.

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